

Vedant Vasant More

Shegaon, Maharashtra, India | 9359811635 | vedantmoremain@gmail.com | [LinkedIn](#) | [GitHub](#)

Professional Summary

AI and Computer Vision-focused BCA student specializing in real-time vision systems, behavioral analysis, and offline-first AI applications. Experienced in building end-to-end AI pipelines using YOLOv8, OpenCV, FastAPI, ONNX Runtime, and speech/vision models under CPU-constrained environments. Strong interest in applied AI research, human behavior understanding, and performance-optimized inference systems. Skilled at architecting privacy-preserving, local-first systems with practical deployment considerations.

Education

Saraswati College, Shegaon

Bachelor of Computer Application (BCA), Artificial Intelligence Focus

Jun 2027

Shegaon

• GPA: 8.25

Technical Skills

- **Programming Languages:** Python, JavaScript, Bash/Shell Scripting, SQL
- **AI/ML and Computer Vision:** YOLOv8, YOLOv8-Pose, OpenCV, PyTorch, ONNX Runtime, Whisper.cpp, InsightFace, ArcFace, MediaPipe, NumPy, ByteTrack, Google Gemini API, Sarvam AI
- **Frameworks and Tools:** FastAPI, Git/GitHub, Uvicorn, Docker, Linux (Arch, Ubuntu, Fedora), Wayland/X11
- **Core Strengths:** Real-time AI inference optimization, Computer vision system design, Multi-stage AI pipeline architecture, Performance optimization under CPU constraints, Behavioral analysis systems, Offline-first and privacy-preserving AI, Technical documentation and system reasoning

Projects

Suspicious AI - Real-Time Behavioral Threat Detection | [GitHub](#)

Nov 2025 - Present

- Architected a real-time behavioral threat detection system capable of distinguishing aggressive activity from friendly interactions using pose-aware reasoning.
- Designed dual-stage system evolution from motion-only detection to pose-aware conflict analysis using 17-keypoint human pose estimation.
- Solved the critical "motion vs. intent" problem by implementing temporal behavior analysis across multiple frames instead of relying on single-frame classification.
- Engineered suppression gates using joint symmetry, separation dynamics, and interaction duration to reduce false positives in hugs and non-aggressive contact.
- Built configurable session-level threat scoring instead of frame-level alerts, improving contextual stability and reducing noise.
- Optimized CPU-only inference pipeline from 7-8 FPS to 20.3 FPS average through ONNX Runtime export, resolution scaling, pipeline threading, and smart frame skipping.
- Reduced false positives by over 80 percent while maintaining real-time responsiveness.
- Demonstrated practical understanding of system architecture, tracking, temporal logic, and behavior-aware AI reasoning.

Smart Attendance System - AI-Powered Face Recognition Attendance | [GitHub](#)

- Built a local-first face recognition attendance system using ArcFace embeddings and cosine similarity matching.
- Architected a complete end-to-end pipeline including face detection, alignment, embedding generation, and attendance management.
- Developed FastAPI backend APIs with browser-based frontend using WebRTC camera access.
- Implemented duplicate prevention, automated check-in/check-out workflow, and attendance history tracking.
- Designed SQLite schema for user records, embeddings, and attendance logs.
- Ensured complete offline functionality with zero external biometric data transfer.

Whisper Type - Offline Voice-to-Text for Linux | [GitHub](#)

Feb 2025 - Feb 2025

- Developed a fully offline voice-to-text assistant compatible with Linux desktop environments and Wayland.
- Integrated Whisper.cpp for local speech recognition without cloud dependency.
- Built dual interaction modes: clipboard copy mode and automatic keyboard injection mode.
- Implemented global hotkey-based activation for system-wide accessibility.
- Designed lightweight GUI and comprehensive multi-distribution setup documentation.

Interview Confidence Detection (Confidential) | [LinkedIn Post](#)

Mar 2026 - Present

- Architected a hybrid multilingual confidence analysis system combining behavioral speech signals with semantic LLM evaluation.
- Built a four-tier architecture integrating transcription, phonation analysis, semantic reasoning, and composite scoring.
- Implemented speech signal processing for jitter, pitch contour, speech rate, silence ratio, and response latency analysis.
- Integrated LLM-based semantic scoring to evaluate relevance, clarity, concreteness, and response depth.
- Achieved 72.8 percent transcription accuracy for Marathi/Hindi audio under real-world conditions.